

Angular Momentum Representation For Free particle

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Abstract

We analyze two indistinguishable fermions in a 1D infinite well with a $g\delta(\mathbf{x}_1 - \mathbf{x}_2)$ contact potential. We demonstrate that the required spatial antisymmetry forces the wavefunction to vanish at particle coincidence ($\Psi(\mathbf{x}, \mathbf{x}) = 0$), rendering the interaction inert. The system is then solved as two non-interacting particles. We derive the position-space eigenfunctions, the momentum-space wavefunctions via Fourier transform, and the two-particle propagator, which takes the form of a Slater determinant.

We establish a general theorem for single-particle operator expectation values, $\langle \hat{A}_1 \rangle = \frac{1}{2}(\langle \hat{a} \rangle_{n_1} + \langle \hat{a} \rangle_{n_2})$, using it to calculate uncertainties and verify the Heisenberg principle. Finally, we construct generalized, $\mathbf{n}$ -dependent ladder operators, showing the system admits a deformed Heisenberg algebra consistent with its non-linear spectrum.

Keywords: Identical Particles, Infinite Potential Well, Contact Interaction

1 Problem Setting

The purpose of this project is to construct a complete and self-contained formulation of the angular-momentum representation for a nonrelativistic free particle. The central objective is to solve the time-dependent Schrödinger equation (TDSE)

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle, \quad \hat{H} = \frac{\hat{\mathbf{p}}^2}{2m},$$

in a basis that diagonalizes simultaneously the free Hamiltonian $\hat{H}$ and the angular-momentum operators $\hat{L}^2$ and $\hat{L}_z$.

The motivation for this construction is threefold:

1. To reveal the role of rotational symmetry in quantum mechanics through the generators of the group $\text{SO}(3)$.
2. To establish a clear connection between spatial symmetries and conservation laws.
3. To demonstrate how angular-momentum eigenstates provide a natural separation of variables for free-particle dynamics in spherical coordinates.

1.1 Physical Scenario

We consider a single spinless particle of mass m moving freely in three-dimensional space. No external forces act on the system, so the Hamiltonian contains only kinetic energy:

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m}.$$

Because the dynamics is invariant under translations and rotations, the corresponding generators $\hat{\mathbf{p}}$ and $\hat{\mathbf{L}}$ play a central role. In particular:

$$[\hat{H}, \hat{\mathbf{L}}] = 0,$$

which implies that $\hat{H}$, $\hat{L}^2$ and $\hat{L}_z$ admit simultaneous eigenstates.

1.2 Goal of the Construction

We aim to:

- Construct the operators $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$ from the canonical commutation relations.
- Express $\hat{\mathbf{L}}$ in both Cartesian and spherical coordinates.
- Establish the commutation relations that define the Lie algebra of $\text{SO}(3)$:

$$[\hat{L}_i, \hat{L}_j] = i\hbar \epsilon_{ijk} \hat{L}_k.$$

- Solve the eigenvalue problems for $\hat{L}_z$ and $\hat{L}^2$, obtaining the spherical harmonics $Y_{\ell m}(\theta, \phi)$.
- Expand the free-particle wavefunction in the basis $|k, \ell, m\rangle$.
- Solve the TDSE in this basis and reconstruct the wavefunction in position, momentum, and energy representations.

1.3 Mathematical Background Assumed

The development assumes familiarity with:

- Hilbert-space formalism: kets, operators, inner products.
- Canonical commutators $[\hat{x}_i, \hat{p}_j] = i\hbar \delta_{ij}$.
- Fourier transforms and the relation between position and momentum representations.

- Basic knowledge of spherical coordinates and special functions (Legendre polynomials).

1.4 Expected Outcome

The final result of this construction is the full decomposition

$$|\psi(t)\rangle = \sum_{\ell, m} \int_0^\infty dk k^2 \psi_{k\ell m}(0) e^{-i\hbar k^2 t/(2m)} |k, \ell, m\rangle,$$

along with the corresponding integral transforms that yield the wavefunction in position-, momentum-, and energy-space representations. This framework encapsulates the full rotational structure of the free particle and sets the stage for generalizations to interacting systems, spin, and relativistic dynamics.

2 start point: Known Results and Preliminary Derivations

This section summarizes—and where necessary proves—the fundamental results that serve as the starting point for the construction of the angular-momentum representation of the free nonrelativistic particle. Although these results are standard, we provide complete derivations to ensure that the document is self-contained.

2.1 Wavefunction Representations for the Free Particle

2.1.1 Position Representation

For a single particle of mass m , the free Hamiltonian is

$$\hat{H} = \frac{\hat{p}^2}{2m}.$$

The time-dependent Schrödinger equation in the position representation is

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{x}, t).$$

To show this, recall that in the position basis,

$$\hat{\mathbf{p}} = -i\hbar \nabla_{\mathbf{x}},$$

so that

$$\frac{\hat{p}^2}{2m} \psi(\mathbf{x}, t) = \frac{1}{2m} (-i\hbar)^2 \nabla^2 \psi(\mathbf{x}, t) = -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{x}, t).$$

2.1.2 Momentum Representation

Define the momentum-space wavefunction

$$\psi(\mathbf{p}, t) = \langle \mathbf{p} | \psi(t) \rangle.$$

Using $\hat{H}|\mathbf{p}\rangle = \frac{p^2}{2m}|\mathbf{p}\rangle$, the Schrödinger equation becomes

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{p}, t) = \frac{p^2}{2m} \psi(\mathbf{p}, t),$$

which integrates to the general solution

$$\psi(\mathbf{p}, t) = \psi(\mathbf{p}, 0) e^{-ip^2 t / (2m\hbar)}.$$

2.1.3 Energy Representation

Energy eigenstates satisfy

$$\hat{H}|E\rangle = E|E\rangle.$$

In the free case $E = p^2/(2m)$, and the time-dependent coefficient satisfies

$$i\hbar \frac{\partial}{\partial t} \psi(E, t) = E \psi(E, t),$$

with the formal solution

$$\psi(E, t) = \psi(E, 0) e^{-iEt/\hbar}.$$

2.2 Integral Transforms Between Representations

2.2.1 Fourier Transform: $\psi(\mathbf{x}, t) \leftrightarrow \psi(\mathbf{p}, t)$

Position and momentum eigenstates satisfy

$$\langle \mathbf{x} | \mathbf{p} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}.$$

To verify orthonormality, compute

$$\begin{aligned} \langle \mathbf{p}' | \mathbf{p} \rangle &= \int d^3x \langle \mathbf{p}' | \mathbf{x} \rangle \langle \mathbf{x} | \mathbf{p} \rangle \\ &= \frac{1}{(2\pi\hbar)^3} \int d^3x e^{-i\mathbf{p}'\cdot\mathbf{x}/\hbar} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \\ &= \delta^{(3)}(\mathbf{p} - \mathbf{p}'). \end{aligned}$$

Thus,

$$\psi(\mathbf{x}, t) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3p e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \psi(\mathbf{p}, t),$$

$$\psi(\mathbf{p}, t) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3x e^{-i\mathbf{p}\cdot\mathbf{x}/\hbar} \psi(\mathbf{x}, t).$$

2.2.2 Transform Between Momentum and Energy Spaces

Let

$$E = \frac{p^2}{2m}.$$

Then for any function $f(p)$ we may write

$$\int_0^\infty dp p^2 f(p) = \int_0^\infty dE \frac{p(E)}{\sqrt{2mE}} (2m) f(\sqrt{2mE}),$$

using the relation

$$\delta\left(E - \frac{p^2}{2m}\right) = \frac{m}{p} \delta(p - \sqrt{2mE}).$$

2.3 Eigenvalue Equations for Basic Operators

2.3.1 Position Operator

Position eigenkets satisfy

$$\hat{\mathbf{x}}|\mathbf{x}\rangle = \mathbf{x}|\mathbf{x}\rangle.$$

Completeness:

$$\int d^3x |\mathbf{x}\rangle\langle\mathbf{x}| = \hat{1}.$$

2.3.2 Momentum Operator

Momentum eigenstates satisfy

$$\hat{\mathbf{p}}|\mathbf{p}\rangle = \mathbf{p}|\mathbf{p}\rangle,$$

and the canonical commutation relation

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$$

implies that in the position representation,

$$\hat{p}_i = -i\hbar \frac{\partial}{\partial x_i}.$$

2.3.3 Free Hamiltonian Eigenstates

Energy eigenstates satisfy

$$\hat{H}|E\rangle = E|E\rangle.$$

Since $\hat{H}$ is a function of $\hat{\mathbf{p}}$, momentum eigenstates are also energy eigenstates:

$$\hat{H}|\mathbf{p}\rangle = \frac{p^2}{2m}|\mathbf{p}\rangle.$$

2.4 Operators of Symmetry: Translations and Time Evolution

2.4.1 Spatial Translations

Define the translation operator

$$\hat{T}(\mathbf{a}) = \exp\left(-\frac{i}{\hbar}\mathbf{a} \cdot \hat{\mathbf{p}}\right).$$

To show that it acts as expected, compute

$$\begin{aligned}\hat{T}(\mathbf{a})|\mathbf{x}\rangle &= \exp\left(-\frac{i}{\hbar}\mathbf{a} \cdot \hat{\mathbf{p}}\right)|\mathbf{x}\rangle, \\ \langle \mathbf{x}' | \hat{T}(\mathbf{a}) | \mathbf{x} \rangle &= \frac{1}{(2\pi\hbar)^3} \int d^3p e^{-i\mathbf{a} \cdot \mathbf{p}/\hbar} e^{i\mathbf{p} \cdot (\mathbf{x}' - \mathbf{x})/\hbar} \\ &= \delta^{(3)}(\mathbf{x}' - (\mathbf{x} + \mathbf{a})),\end{aligned}$$

so that

$$\hat{T}(\mathbf{a})|\mathbf{x}\rangle = |\mathbf{x} + \mathbf{a}\rangle.$$

2.4.2 Time Evolution

The unitary time-evolution operator is

$$\hat{U}(t) = \exp\left(-\frac{i}{\hbar}\hat{H}t\right).$$

Expanding for infinitesimal δt ,

$$\hat{U}(\delta t) = 1 - \frac{i}{\hbar}\hat{H}\delta t + O(\delta t^2).$$

Acting on a state $|\psi(t)\rangle$ gives

$$|\psi(t + \delta t)\rangle = \left(1 - \frac{i}{\hbar}\hat{H}\delta t\right)|\psi(t)\rangle,$$

which leads directly to the Schrödinger equation.

2.5 Correspondence Principle and Ehrenfest's Theorem

2.5.1 General Ehrenfest Theorem

Let $\hat{A} = \hat{A}(t)$ be an operator (possibly explicitly time-dependent) with expectation value

$$\langle \hat{A} \rangle_t = \langle \psi(t) | \hat{A}(t) | \psi(t) \rangle.$$

Differentiate with respect to time:

$$\frac{d}{dt}\langle\hat{A}\rangle_t = \left\langle\frac{\partial\hat{A}}{\partial t}\right\rangle_t + \left\langle\frac{d\hat{A}}{dt}\right\rangle_{\text{state}}.$$

The contribution from the time evolution of the state is

$$\left\langle\frac{d\hat{A}}{dt}\right\rangle_{\text{state}} = \left\langle\frac{1}{i\hbar}[\hat{A}, \hat{H}]\right\rangle_t,$$

which follows from using the Schrödinger equation $i\hbar\partial_t|\psi\rangle = \hat{H}|\psi\rangle$ and its adjoint. Combining both contributions yields the Ehrenfest theorem:

$$\boxed{\frac{d}{dt}\langle\hat{A}\rangle_t = \frac{1}{i\hbar}\langle[\hat{A}, \hat{H}]\rangle_t + \left\langle\frac{\partial\hat{A}}{\partial t}\right\rangle_t}.$$

Proof.

Compute directly:

$$\begin{aligned}\frac{d}{dt}\langle\hat{A}\rangle_t &= \frac{d}{dt}\langle\psi|\hat{A}|\psi\rangle \\ &= \langle\dot{\psi}|\hat{A}|\psi\rangle + \langle\psi|\dot{\hat{A}}|\psi\rangle + \langle\psi|\hat{A}|\dot{\psi}\rangle.\end{aligned}$$

Using $i\hbar|\dot{\psi}\rangle = \hat{H}|\psi\rangle$ and its adjoint $-i\hbar\langle\dot{\psi}| = \langle\psi|\hat{H}$, we have

$$\langle\dot{\psi}|\hat{A}|\psi\rangle + \langle\psi|\hat{A}|\dot{\psi}\rangle = \frac{1}{i\hbar}\langle\psi|[\hat{A}, \hat{H}]|\psi\rangle.$$

Hence the result.

2.5.2 Application to Position and Momentum

Let $\hat{A} = \hat{\mathbf{x}}$. Since $\partial\hat{\mathbf{x}}/\partial t = 0$ (in the Schrödinger picture),

$$\frac{d}{dt}\langle\hat{\mathbf{x}}\rangle_t = \frac{1}{i\hbar}\langle[\hat{\mathbf{x}}, \hat{H}]\rangle_t.$$

For the free Hamiltonian $\hat{H} = \hat{\mathbf{p}}^2/(2m)$, using the canonical commutator $[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$,

$$[\hat{x}_i, \hat{H}] = \frac{1}{2m}([\hat{x}_i, \hat{p}_j]\hat{p}_j + \hat{p}_j[\hat{x}_i, \hat{p}_j]) = \frac{i\hbar}{m}\hat{p}_i,$$

so that

$$\frac{d}{dt}\langle\hat{\mathbf{x}}\rangle_t = \frac{\langle\hat{\mathbf{p}}\rangle_t}{m}.$$

Now take $\hat{A} = \hat{\mathbf{p}}$. Again $\partial\hat{\mathbf{p}}/\partial t = 0$ in the Schrödinger picture and

$$\frac{d}{dt}\langle\hat{\mathbf{p}}\rangle_t = \frac{1}{i\hbar}\langle[\hat{\mathbf{p}}, \hat{H}]\rangle_t.$$

For a free particle $[\hat{\mathbf{p}}, \hat{H}] = 0$, hence

$$\frac{d}{dt}\langle\hat{\mathbf{p}}\rangle_t = 0.$$

Combining both results yields

$$m\frac{d^2}{dt^2}\langle\hat{\mathbf{x}}\rangle_t = 0,$$

which is precisely the classical equation of motion for a free particle. This exemplifies the correspondence principle: expectation values of quantum observables follow classical equations when the operators are identified with classical dynamical variables and when fluctuations are negligible.

2.5.3 Ehrenfest Theorem for a General Potential

For completeness, consider a particle subject to a scalar potential $V(\hat{\mathbf{x}})$. Then $\hat{H} = \hat{\mathbf{p}}^2/(2m) + V(\hat{\mathbf{x}})$. The commutator

$$[\hat{p}_i, V(\hat{\mathbf{x}})] = -i\hbar\frac{\partial V}{\partial x_i}(\hat{\mathbf{x}}),$$

so Ehrenfest's theorem gives

$$\frac{d}{dt}\langle\hat{p}_i\rangle_t = -\left\langle\frac{\partial V}{\partial x_i}(\hat{\mathbf{x}})\right\rangle_t.$$

Thus

$$m\frac{d^2}{dt^2}\langle\hat{\mathbf{x}}\rangle_t = -\langle\nabla V(\hat{\mathbf{x}})\rangle_t,$$

which mirrors Newton's second law, $m\ddot{\mathbf{x}} = -\nabla V(\mathbf{x})$, up to the replacement of forces by expectation values.

2.5.4 Discussion: Correspondence Principle

The correspondence principle, in the present context, asserts that the quantum-mechanical expectation values obey equations that reproduce classical mechanics in the limit where quantum fluctuations around the expectation values are negligible (for example, for sufficiently localized wave packets with small spread compared to scale variations of the potential). The Ehrenfest relations derived above make this statement precise: quantum expectation values obey the classical equations of motion with the classical variables replaced by corresponding expectation values. Deviations

from classical trajectories arise from the terms involving higher moments (fluctuations) of the quantum state, which are not captured by a closed set of equations for the first moments alone unless the potential is at most quadratic.

Remark. The Ehrenfest theorem does not imply that individual quantum particles follow deterministic classical trajectories; it only guarantees that the first moments of suitably prepared states evolve classically under the conditions discussed.